MADHANPRASATH S

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OBJECTIVE

To obtain a challenging position in a progressive organization where I can apply my technical skills and theoretical knowledge gained through my Computer Science and Engineering degree to contribute to innovative projects, enhance my learning, and grow as a software development professional.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript (ES6), C, C++, Haskell

Web Development: HTML5, CSS3, React.js, Express.js, Node.js, REST API Development

Backend & DB: MongoDB, Mongoose, SQL, Firebase, JWT Authentication

ML & AI: TensorFlow, Keras, scikit-learn, NumPy, Pandas, Flask

Tools: Git, GitHub, GitLab, Docker, Postman, VS Code, Linux/Unix

CS Core: DSA, OOP, Operating Systems, DBMS, Computer Networks

Other: Problem Solving, Agile, CI/CD Basics, AWS Basics

EXPERIENCE

Full Stack Web Development Intern *05 July 2024 – 19 July 2024*

CT Infotech India Private Limited — Karur, Tamil Nadu

- **Completed an academic internship specializing in Full Stack Web Development as part of the Manufacturing Data Lake software product development project.**
- **Gained hands-on experience building and integrating web-based interfaces and backend services in a professional software development environment.**
- **Applied industry-standard tools and workflows for data-driven manufacturing applications using HTML, CSS, JavaScript, and related web technologies.**

ACADEMIC PROJECTS

Digital Learning Platform for Rural School Students (NabhaEdu) *2025 – 2026*

- **Built an offline-first PWA using React Native and Firebase for rural school students in Nabha, Punjab with role-based dashboards for Students, Teachers, Parents, and Admin.**
- **Developed modules for lessons, quizzes, attendance, and real-time progress tracking; enabled offline learning via AsyncStorage with Firebase Realtime Database sync.**
- **Added multilingual support (English & Punjabi); published in JSTAR, Vol. 7, Issue 1, 2026.**

Crop Disease Detection using Machine Learning

- **Built a CNN-based image classification model using TensorFlow and Keras to detect and classify crop diseases from leaf images, achieving over 90% validation accuracy on the PlantVillage dataset.**
- **Developed a Flask REST API to serve the trained model, allowing farmers to upload leaf images via a web interface and receive instant disease diagnosis with recommended treatments.**
- **Applied data augmentation techniques (rotation, flipping, zoom) to improve model generalization across varying lighting and environmental conditions.**

Crop Livestock Management System

- **Developed a full-stack web platform enabling farmers to log crop symptoms and receive disease recommendations, reducing dependency on agronomists for routine diagnosis.**
- **Integrated a livestock health tracker for managing animal health records, vaccination schedules, and feeding patterns for improved herd management.**
- **Designed a real-time monitoring dashboard aggregating crop and livestock status, built with Node.js and MongoDB for scalable data handling.**

EDUCATION

B.E – Computer Science and Engineering *2022 – 2026*

Hindusthan Institute of Technology, Coimbatore **CGPA: 7.49**

Coursework: Computer Networks, DBMS, OOP, Operating Systems

CERTIFICATIONS

- **Machine Learning – Infosys Springboard**
- **Introduction to IoT – NPTEL**
- **MongoDB Introduction – MongoDB University**